Phase B - From Unit Distance and Sum-Product to Polynomial Image Compression

A quantitative framework for deciding when a two-variable polynomial can have an unusually small image on a finite set.

Cycle verdict
Five distinct abstraction candidates were extracted, formalized, tested, and audited. Four survive as conditional quantitative principles. The unqualified collision-energy analogy fails. The current bottleneck is rank-two action-carrier mixing. Another Phase B cycle is recommended; Phase C construction experiments are not yet justified.

Scope discipline

This report does not modify the frozen Phase A document, does not claim a new extremal construction, and does not enter Phase C. It uses the supplied unit-distance chain-of-thought only as discovery history, never as proof authority.

Independent study notes - 26 August 2026

How to read this report

Epistemic labels

Label	Meaning
SOURCE	Directly stated or proved in a verified supplied source.
BACKGROUND	Standard mathematics or an elementary derivation used to make the framework explicit.
ABSTRACTED PRINCIPLE	A quantitative principle extracted from source mechanisms and proved here at the stated level.
TRANSFER HYPOTHESIS	A proposed extension to general polynomial images; not silently promoted to a theorem.
EXTERNAL RESULT	A result outside the supplied source set. No external result is invoked in this cycle.
FAILURE	A proposed transfer that is false, logically insufficient, or quantitatively useless.

Source hierarchy actually used

Tier	Document	Use
Primary	Phase_A_Sum_Product_Reproduction_FINAL_FROZEN.pdf	The direct A = GP construction, directness, bounded distortion, and rate ledger.
Primary	Planar Point Sets with Many Unit Distances	Proposition 2.2, Theorem 2.3, and Lemmas 2.4-2.6.
Supporting	Remarks on the Disproof of the Unit Distance Conjecture	Human-digested simplification and interpretation.
Supporting	The Unit Distance Problem and the Sum-Product Phenomenon: Final Frozen Guide	Source-disciplined bridge and warnings about the heuristic routes.
Discovery only	unit-distance-cot.pdf	Not used as proof authority.

Exactly five candidates

ID	Candidate	Question	Audit result
B1	Collision energy	Is large representation multiplicity the common core?	TOO WEAK without support control
B2	Hybrid action/carrier	When does A = structured action x dense carrier survive inside f?	VALID sufficient principle
B3	Coordinate linearization	Can a change of variables expose a one-dimensional group law?	VALID sufficient principle
B4	Bounded distortion	Can all polynomial outputs be placed in a small countable region?	VALID with an exponent test
B5	High-dimensional amplification	When does a per-coordinate gain become a power saving?	VALID rate lemma

1. The collision formulation

The mandatory starting point and its exact logical direction

[BACKGROUND] Let A be finite, $N = |A|$, and let f be a two-variable polynomial. Define

$$r_f(z) = \#\{(a,b) \text{ in } A^2 : f(a,b)=z\},$$

$$E_f(A) = \#\{(a,b,c,d) \text{ in } A^4 : f(a,b)=f(c,d)\} = \Sigma_z\, r_f(z)^2.$$

Since $\Sigma_z\, r_f(z)=N^2$, Cauchy-Schwarz gives

$$N^4 \le |f(A,A)|\, E_f(A), \text{ hence } |f(A,A)| \ge N^4/E_f(A).$$

First correction
A small image forces large collision multiplicity. The displayed inequality does not say that a large collision count forces a small image. Its direction is a lower bound for the image.

In particular, a target $|f(A,A)| \le N^{2-\delta}$ requires

$$E_f(A) \ge N^{2+\delta}.$$

This is a necessary threshold and a useful rejection test. It is not a construction certificate.

Why energy alone cannot control support

[FAILURE] Consider an abstract map on N^2 inputs with one fiber of size about $N^2/2$ and about $N^2/2$ singleton fibers. Its collision energy is at least $N^4/4$, yet its image still has order N^2. A second moment can be dominated by a small heavy part while a long light tail keeps the support large.

Two valid support certificates

[ABSTRACTED PRINCIPLE] Uniform fiber coverage. If every occupied value satisfies $r_f(z) \ge m$, then $|f(A,A)| \le N^2/m$.

[ABSTRACTED PRINCIPLE] Tail-complete container. If all but T input pairs map into a set C, then $|f(A,A)| \le |C|+T$. Each exceptional pair can create at most one additional output.

These statements reveal the correct division of labor: the collision equation diagnoses possible structure, while a support or coverage argument proves compression.

2. First source bridge: unit distance

The exact restricted-difference identity

[SOURCE] For a finite planar set P identified with a subset of the complex plane, define $r_P(u)=\#\{(x,y) \text{ in } P^2: y-x=u\}$. Then

$$2\nu(P) = \Sigma_{|u|=1}\, r_P(u).$$

Thus unit distances are the mass of the additive difference representation function restricted to the unit circle.

What the successful construction actually supplies

[SOURCE] Proposition 2.2 constructs a set U of distinct bounded-denominator norm-one elements with $|U| \geq e^{\gamma f}$. Every complex embedding of each u in U has modulus one.

[SOURCE] Lemma 2.4 chooses a lattice coset window X for which the aggregate number E_a of ordered pairs (x,x+u), u in U, obeys $E_a \geq e^{\gamma f/2}|X|$. The proof obtains this from torus averaging and the product-of-discs overlap factor $\rho_R^{\,f}$.

[SOURCE] Lemmas 2.5-2.6 project injectively to the plane and bound $|P| \leq e^{Bf}$. Theorem 2.3 converts the excess rate $\gamma/2$ into $\nu(P) \geq |P|^{1+\delta}$.

The transferable pattern, with the quantifiers exposed

algebraic supply of selected outputs U + aggregate realization $\Sigma_{u \text{ in } U}\, r_X(u)$.

[ABSTRACTED PRINCIPLE] For a lower-bound problem about a selected relation, both the size of the allowed output family and its aggregate realization count matter. The source proof does not require every u in U to be individually popular in the chosen window.

> **Where the analogy reverses**
> Unit distance benefits from many selected displacement values u because the goal is to count a relation. Polynomial-image compression benefits from few total output values. The shared object is representation accounting; the sign of the support objective is different.

[FAILURE] It is therefore incorrect to identify the unit-distance construction with an energy proof of a small image. The source controls a first moment over a selected set U, not the full second moment of all differences, and its conclusion is a large edge count rather than a small support.

3. Second source bridge: the frozen A = GP construction

Verified ingredients

[SOURCE] In a totally real field K of degree d, Phase A takes an additive box $P=X+B^{+}(\varepsilon X)$, a logarithmic unit box $G=B^{\times}(Y)$, and $A=GP$.

- Directness: the map G x P -> GP is injective, so $|A|=|G||P|$.
- Bounded distortion: $e^{-Y} \le |\sigma_i(g)| \le e^{Y}$ for every g in G and every real embedding.
- Product compression: $|AA|/|A|^2 \le |GG|/|G|^2 \le c_0^{-d} Y^{1-d} \Delta_K^{\,2}$.
- Additive container: $|A+A|/|A| \le (e^{Y}/c_0)^{d} \Delta_K^{\,1/2}$.
- High-degree conversion: bounded root discriminant controls the costs, a fixed Y creates a per-degree product saving, X pays additive spread, and d converts the rate into a power of |A|.

What is genuinely collision-theoretic

Small |GG| implies large multiplicative energy of G by Cauchy-Schwarz, but the source proof does not use energy as an upper-support theorem. It proves a direct support containment:

$$AA = (GG)(PP), \text{ so } |AA| \le |GG||P|^2.$$

Directness then makes the normalization exact:

$$|AA|/|A|^2 \le |GG|/|G|^2.$$

[ABSTRACTED PRINCIPLE] The transferable core is not merely 'many product collisions.' It is a factor-through support bound plus controlled parameter multiplicity.

Two indispensable bridges

Source mechanism	Abstract form	Cost if weakened
$\|GP\|=\|G\|\|P\|$	A parameterization has fiber size at most μ	A normalized two-input bound loses μ^2
$\|\sigma_i(g)\| \le e^{Y}$	The action keeps all outputs in a controlled carrier coordinate region	The output-container exponent grows by the distortion rate

4. Reusable decision framework

For a polynomial f, the framework now asks the following questions in order. This ordering prevents a collision calculation from being mistaken for a compression proof.

	Stage	Required output
1	Normalize	Fix the coefficient field, domain, degeneracies, and the target exponent.
2	Find exact normal forms	Test affine shifts, power maps, additive and multiplicative compositions, norms, and traces. Record fibers.
3	Write the collision equation	Factor f(a,b)-f(c,d); identify symmetries, ratios, group laws, and low-dimensional invariants.
4	Choose a mechanism	Normal-form set, hybrid action/carrier, or arithmetic output container.
5	Audit directness	Bound the multiplicity of every parameterization and coordinate map.
6	Prove support control	Give an exact factor-through bound, a complete container, or uniform fiber coverage.
7	Write the ledger	Convert every gain and cost into a per-degree or per-coordinate exponent.
8	Stress test	Check light tails, operation mismatch, coefficient growth, low-dimensional artifacts, and exceptional normal forms.
9	Classify	VALID, PROMISING BUT UNPROVED, TOO WEAK, or FALSE.

Decision rule

Do not advance a candidate merely because its collision equation looks structured. Advance it only after the structure yields a support bound and the net gain exponent beats every distortion, multiplicity, height, and covolume cost.

5. Candidate B1 - Collision-energy abstraction

EXTRACT

[SOURCE] The unit-distance proof constructs many selected norm-one translations and a window with large aggregate translation overlap. The Phase A product proof constructs a small product support GG and transfers that support saving to AA by directness.

FORMALIZE

[TRANSFER HYPOTHESIS] Both proofs might be instances of 'engineer large representation multiplicity, then infer an extremal conclusion.'

For polynomial images the natural statistic is $E_f(A)$, and the target $N^{2-\delta}$ requires $E_f(A) \geq N^{2+\delta}$.

TEST

Polynomial	Collision statistic	What the test shows
x+y	$E_f=E^+(A)$	Small additive doubling gives E_f of order at least N^3/K.
xy	$E_f=E^x(A)$	Small product set gives E_f of order at least N^3/K.
x+y+xy	$E_f=E^x(A+1)$	Affine shift exposes the right energy.
x^2+y^2	Energy of representations by A^2+A^2	The transformed carrier, not A itself, must be structured.
xy(x+y)	Only symmetry is immediate	On a geometric progression the energy is $2N^2-N$, far below a compression threshold.

AUDIT

- Does it restate many collisions? Yes, unless it identifies a support or coverage mechanism.
- Does large energy imply the desired image estimate? No; Cauchy-Schwarz points in the opposite logical direction.
- Does the unit-distance source prove a second-moment statement? No; it controls a selected first moment.

UPDATE AND MUTATE

[FAILURE] Unqualified collision energy is TOO WEAK.

[ABSTRACTED PRINCIPLE] Retain the collision equation as a diagnostic and necessary threshold. Mutate the target from energy alone to support control plus fiber coverage. This directly attacks the light-tail failure.

6. Candidate B2 - Hybrid carrier/action abstraction

EXTRACT

[SOURCE] Phase A uses A=GP. G carries the nontrivial product compression, P supplies most of the cardinality, directness prevents a collision divisor, and bounded distortion keeps additive outputs in one box.

FORMALIZE: factor-through hybrid lemma

[ABSTRACTED PRINCIPLE] Let A be the image of $\alpha: S \times P \to A$, and suppose every fiber of α has size at most μ. Suppose there are maps q, r, and Ψ such that

$$f(\alpha(s_1,p_1), \alpha(s_2,p_2)) = \Psi(q(s_1,s_2), r(p_1,p_2)).$$

Then $|A| \geq |S||P|/\mu$ and

$$|f(A,A)|/|A|^2 \leq \mu^2 \, (|q(S,S)|/|S|^2) \, (|r(P,P)|/|P|^2).$$

The proof is an exact support count: the output is determined by one q-value and one r-value.

TEST

- For $f=xy$ and $\alpha(g,p)=gp$, take $q(g_1,g_2)=g_1g_2$, $r(p_1,p_2)=p_1p_2$, and $\Psi(q,r)=qr$. This recovers the Phase A product estimate.
- For $f=x+y$, the expression $g_1p_1+g_2p_2$ has no rank-one separation of this type. The source uses a geometric container instead.
- For $f=x+y+xy$, first shift $B=A+1$; the resulting product form is rank one.
- For $f=x^2+y^2$, independent action parameters occur in an additive sum; transformed-carrier structure is more natural.
- For $h=xy(x+y)$, two coupled separable terms remain. Exact rank-one factorization fails in the native hybrid variables.

AUDIT

- Directness is not cosmetic: $\mu=e^{\kappa d}$ costs $e^{2\kappa d}$ after normalization.
- A decomposition of the formula is useful only if the q and r supports are quantitatively small.
- The principle is sufficient, not universal; it makes no claim that every polynomial respects a hybrid decomposition.

UPDATE AND MUTATE

[ABSTRACTED PRINCIPLE] The factor-through hybrid lemma is VALID.

[TRANSFER HYPOTHESIS] For separation rank at least two, replace exact factorization by a correlated-sum problem. No saving is currently verified.

7. Candidate B3 - Coordinate-linearization abstraction

EXTRACT

[SOURCE] Phase A uses ordinary embeddings to linearize addition and logarithmic embeddings to linearize multiplication. The same objects are useful only because bounded logarithmic coordinates control their action in ordinary coordinates.

FORMALIZE: coordinate normal-form lemma

[ABSTRACTED PRINCIPLE] Let star be a one-dimensional group operation. If

$$f(x,y)=F(G(x) \text{ star } H(y)),$$

then

$$|f(A,A)| \le |G(A) \text{ star } H(A)|.$$

If G and H have fibers of size at most μ_G and μ_H on A, then $|G(A)| \ge N/\mu_G$ and $|H(A)| \ge N/\mu_H$. This bookkeeping separates genuine group-law compression from collapse caused by the coordinate map.

TEST: exact normal forms

Polynomial	Coordinate package	Audit
x+y	G(x)=x, H(y)=y, star=+	Native additive coordinate
xy	G(x)=x, H(y)=y, star=x	Native multiplicative coordinate; log is optional on positive sets
x+y+xy	G(x)=x+1, H(y)=y+1, star=x, F(t)=t-1	Affine shift converts the formula exactly to a product
x^2+y^2	G(x)=x^2, H(y)=y^2, star=+	Squaring has fiber at most two over the reals
xy(x+y)	No one-dimensional form exhibited	Native additive and multiplicative charts simplify different factors but not the whole polynomial

AUDIT

- A logarithm requires a positive domain and turns multiplication into addition but does not simplify an added term.
- A noninjective chart must report its fiber cost.
- No theorem is invoked saying the displayed additive and multiplicative forms exhaust all relevant polynomials.

UPDATE AND MUTATE

[ABSTRACTED PRINCIPLE] Coordinate normal forms are a VALID sufficient branch of the decision framework.

[TRANSFER HYPOTHESIS] For polynomials with no exhibited one-dimensional form, introduce separation rank after allowed bounded-fiber coordinate changes, then test whether a rank-two correlated-sum estimate is possible.

8. Candidate B4 - Bounded-distortion abstraction

EXTRACT

[SOURCE] In unit distance, every selected translation has exact modulus one in every embedding, so each product-disc overlap is the same. In Phase A, every g in G satisfies $|\sigma_i(g)| \le e^Y$, so A+A remains in one ordinary Minkowski box.

FORMALIZE: arithmetic output-container lemma

[BACKGROUND] Let K be totally real of degree d, let A be contained in the ring of integers, and assume $|\sigma_k(a)| \le M$ for every a in A and every embedding. For f in Z[x,y] of total degree m, write $H_1(f)$ for the sum of the absolute values of its coefficients. Then every output is an algebraic integer and

$$|\sigma_k(f(a,b))| \le H_1(f)\ \max(1,M)^m.$$

The additive lattice packing estimate used in Phase A therefore gives

$$|f(A,A)| \le (2H_1(f)\ \max(1,M)^m + 1)^d.$$

[ABSTRACTED PRINCIPLE] Bounded distortion becomes useful only after the output region is counted and compared with $|A|^2$. 'Bounded' without a rate is not a mathematical certificate.

TEST ON THE PHASE A HYBRID

For A=GP, the source gives $M \le 2Xe^Y$. The generic polynomial container has per-degree logarithmic cost approximately

$$m(\log X + Y) + O_f(1).$$

The two-input budget is approximately twice the per-degree logarithm of |A|. Consequently:

- For linear outputs, the X-exponent is favorable; this is why the additive container can be near-linear.
- For degree two, the X-exponent already consumes the quadratic budget before regulator, discriminant, and e^{Yd} costs are paid.
- For degree three, the universal container is even less competitive.
- The exact shifted-product form of x+y+xy and the squared-additive form of x^2+y^2 beat the generic degree bound because they are found before the container step.

AUDIT, UPDATE, MUTATE

[FAILURE] The statement 'bounded coordinates compress every fixed-degree polynomial' is TOO WEAK.

[ABSTRACTED PRINCIPLE] The quantified output-container lemma is VALID as a fallback. Mutate the degree cost by first removing every exactly separable component of f; only the residual polynomial should be charged to the generic box.

9. Candidate B5 - High-dimensional amplification

EXTRACT

[SOURCE] Unit distance compares $|U| \geq e^{\gamma f}$, overlap cost ρ_R^f, and $|P| \leq e^{Bf}$. Phase A compares a normalized product gain e^{-sd} with $|A| \leq e^{Bd}$.

FORMALIZE: two rate-conversion lemmas

[ABSTRACTED PRINCIPLE] Normalized image form. If $|A_d| \leq e^{Bd}$ and $|f(A_d, A_d)| \leq e^{-sd}|A_d|^2$ with $s>0$, then

$$|f(A_d, A_d)| \leq |A_d|^{2-s/B}.$$

[ABSTRACTED PRINCIPLE] Absolute container form. If $|A_d| \geq e^{ad-o(d)}$ and $|f(A_d, A_d)| \leq e^{cd+o(d)}$, then

$$|f(A_d, A_d)| \leq |A_d|^{c/a+o(1)}.$$

A quadratic power saving requires $c<2a$.

Unified gain-versus-cost ledger

Setting	Gain	Costs	Required sign
Unit distance	Direction supply γ	Window overlap $-\log \rho_R$, packing rate B	$\gamma+\log \rho_R$ must be positive
Phase A product	Log-box compression s	Regulator, discriminant, constants, multiplicity	Net normalized rate $s_{net}>0$
Polynomial container	Carrier density $2a$	Degree, height, distortion, covolume	$2a-c>0$
Bounded multiplicity	Source saving s	2κ when $\mu_d \leq e^{\kappa d}$	$s-2\kappa>0$

AUDIT

- Dimension amplifies all constants and all covolume penalties, not only the desired gain.
- The order of parameters matters because one parameter can improve one rate and worsen another.
- A fixed-dimensional saving factor is only a constant; a power saving needs d comparable to log |A|.

UPDATE

[ABSTRACTED PRINCIPLE] High-dimensional amplification is VALID only as a rate-conversion lemma after a strictly positive net rate has already been proved.

[FAILURE] The slogan 'high dimension helps' is FALSE without the signed ledger.

10. Collision library for the basic polynomial tests

Polynomial	Collision equation	Hidden operation	Natural carrier	Cycle verdict
x+y	a+b=c+d	Native addition	Arithmetic progression or additive box	Applies; $E \geq N^3/K$ when $\|A+A\| \leq KN$
xy	ab=cd	Native multiplication	Geometric progression or Phase A hybrid	Applies; direct support control is available
x+y+xy	(a+1)(b+1)=(c+1)(d+1)	Affine-shifted multiplication	A=B-1 with B multiplicatively structured	Exact normal form
x^2+y^2	$a^2+b^2=c^2+d^2$	Addition on A^2	A chosen so A^2 is additively structured	Works after a fiber-at-most-two map
xy(x+y)	ab(a+b)=cd(c+d)	Rank-two hybrid expression	Geometric progression stress test	Fails there: image N(N+1)/2

The mixed stress test in detail

[BACKGROUND] Take h(x,y)=xy(x+y) and $A=\{q^0,\ldots,q^{N-1}\}$ with an integer $q \geq 3$. For i<j,

$$h(q^i,q^j) = q^{2i+j} + q^{i+2j}.$$

Its base-q expansion has two digits equal to one in distinct positions. Those two positions recover i and j. When i=j, the output is $2q^{3i}$, whose single digit two is also unique because $q \geq 3$. Therefore the output determines the unordered pair {i,j}.

$$|h(A,A)|=N(N+1)/2,\ E_h(A)=N+4(N \text{ choose } 2)=2N^2-N.$$

[FAILURE] This proves that excellent multiplicative structure of A need not compress a polynomial containing an uncontrolled additive factor.

11. Phase B mechanism table

Unit-distance ingredient	Sum-product analogue	General polynomial principle
Distinct norm-one U from split-prime choices	Dense logarithmic unit box G with small GG	Construct an output invariant family and count its distinct values
Aggregate overlap $\Sigma_{u \text{ in } U} r_X(u)$	Dense carrier P and direct support containment	Keep output supply, realization multiplicity, and total support as separate quantities
Fixed modulus in every embedding	Bounded $\lvert\sigma_i(g)\rvert \le e^Y$	Quantified distortion in the coordinate system that counts outputs
Injective coordinate projection	Injective GP parameterization and real field embedding	Bound every projection and parameterization fiber
Minkowski product-disc window	Ordinary Minkowski additive box	Tail-complete output container with a lattice-point bound
e^{Yf} gain versus e^{Bf} point budget	e^{-sd} normalized saving versus e^{Bd} size budget	Signed exponential-rate ledger and power conversion
All embeddings retained before planar projection	Ordinary and logarithmic embeddings used together	A coordinate package must state what each chart simplifies and how charts interact

[ABSTRACTED PRINCIPLE] The common architecture is: algebraic parameter supply, a carrier on which the parameters act, exact or bounded-multiplicity transfer to the target, a countable output region, and a positive gain-minus-cost rate. The extremal statistic being optimized is not the same in the two source problems.

12. Adversarial audit of all five candidates

Audit question	Finding	Required safeguard
1. Mere restatement?	B1 failed until support control was added.	Require a container or full fiber coverage.
2. Noninjective coordinate?	B3 tracks bounded fibers; $x \to x^2$ costs at most two over R.	Record domain and fiber bound.
3. Operation preserved?	Affine shift preserves the shifted product identity; log does not simplify sums.	Write the exact identity before counting.
4. Multiplication destroyed by addition?	Yes for $xy(x+y)$ on a geometric progression.	Move to separation-rank analysis.
5. Addition destroyed by multiplication?	Yes for generic A=GP inside x+y.	Use a bounded-distortion additive container.
6. Low-dimensional artifact?	B5 explicitly requires a uniform positive rate as d grows.	No conclusion from a fixed d constant.
7. Exceptional normal form?	$x+y+xy$ and x^2+y^2 reduce exactly.	Test normal forms before generic estimates.
8. Height/degree cancels gain?	Generic degree-m container costs M^{md}.	Put degree and coefficient height in the ledger.
9. Energy implies image?	No; the Cauchy-Schwarz direction was corrected.	Use energy only as necessary evidence.
10. Genuinely broader?	B2-B5 are stated as quantitative lemmas independent of the original notation.	Generic application remains a transfer hypothesis.

Final classifications

Candidate	Classification	Surviving content
B1	TOO WEAK, then mutated	Collision threshold plus tail-complete support certificates
B2	VALID	Factor-through hybrid lemma with μ^2 cost
B3	VALID	Bounded-fiber coordinate normal forms as sufficient conditions
B4	VALID but often quantitatively weak	Arithmetic output-container lemma and degree-cost test
B5	VALID	Signed rate-conversion lemma

13. Validated abstraction library

Principle	Content	Quantitative form
Collision-support	Small image requires large E_f, but proof of small image requires support or coverage control.	$N^4 \le \lvert f(A,A)\rvert E_f$
Hybrid factor-through	Separate action and carrier statistics through Ψ; pay parameter multiplicity.	ratio $\le \mu^2 k_S k_P$
Coordinate normal form	Reduce f to one group law under bounded-fiber maps.	$\lvert f(A,A)\rvert \le \lvert G(A) \text{ star } H(A)\rvert$
Output container	Bound every conjugate, preserve integrality, count lattice points.	$(2H_1 M^m+1)^d$
Rate amplification	Convert a positive net exponential rate into a power of |A|.	$e^{-sd}\lvert A\rvert^2 \le \lvert A\rvert^{2-s/B}$

Minimal valid certificate for small polynomial image

A candidate must provide (i) a structural factorization or complete output container, (ii) controlled fibers for every map used to build A or change coordinates, and (iii) a positive signed rate after all costs. Collision energy alone is not on this list.

Failure library

ID	Rejected principle	Precise failure	Mutation
F1	Large collision energy is sufficient	Wrong inequality direction; heavy fibers may coexist with a long singleton tail	Add support/tail control
F2	A=GP survives every polynomial	Independent G and P variables couple across sums of monomials	Separation rank and correlated sums
F3	Any simplifying coordinate is legitimate	Unbounded fibers can fake compression; domains may be invalid	Fiber and domain ledger
F4	Bounded coordinates compress fixed degree	Degree m causes M^{md} output spread	Remove exact components first
F5	Growing degree automatically helps	Costs are exponentiated too	Require positive net rate

14. Current bottleneck and open lemmas

The bottleneck: rank-two action-carrier mixing

For $\alpha(g,p)=gp$ and $f(x,y)=\Sigma\, c_{ij}x^iy^j$, substitution always gives a finite separable expansion

$$f(g_1p_1,g_2p_2) = \Sigma_{(i,j)}\, c_{ij}(g_1{}^i g_2{}^j)(p_1{}^i p_2{}^j).$$

Rank one reproduces exact factor-through compression. At rank at least two, the trivial inequality for a sum of component supports multiplies their sizes and usually destroys the exponent. The shared parameters may create correlations, but none of the supplied source mechanisms bounds them.

Precise research question

Give checkable hypotheses on U_1,U_2,V_1,V_2 for which the shared-parameter set $\{U_1(g)V_1(p)+U_2(g)V_2(p)\}$ has size at most $e^{-sd}|G|^2|P|^2$ for some $s>0$. The test $h(x,y)=xy(x+y)$ is the first concrete obstruction.

Open-lemma register

ID	Statement	Status	Consequence
O1	Rank-two correlated-sum lemma	UNKNOWN	Would extend the hybrid principle beyond exact factorization
O2	Collision regularization with a controlled light tail	FALSE without extra hypotheses	Would make energy a sufficient support tool
O3	Quantitative bounded-fiber coordinate-package criterion	UNVERIFIED	Would formalize the normal-form decision branch
O4	Residual-degree output container after partial separation	UNKNOWN	Would reduce the generic M^{md} cost
O5	Net-rate stability under $\mu_d \le e^{\kappa d}$	BOOKKEEPING KNOWN	A source rate s survives only if $s>2\kappa$

15. End-of-cycle recommendation

Recommendation: another Phase B abstraction cycle
The framework is not ready for Phase C. The next cycle should study separation rank, correlated sums of two separable terms, and tail-complete support control. It should keep the same five-step discipline and begin with xy(x+y) as an adversarial benchmark, not as a proposed construction target.

Why not external literature verification yet?

The immediate gap is internal: the framework needs a mathematically sharp statement of what replaces exact rank-one factorization. Literature verification becomes efficient only after that statement identifies the relevant object, such as a correlated sum of monomial-feature sets or a bounded-fiber normal-form class.

Why not Phase C?

The source mechanisms currently give rigorous compression only for exact group-law forms or for outputs whose full arithmetic container beats the quadratic budget. The first genuinely mixed polynomial defeats the naive multiplicative test. Starting construction experiments now would search without a validated transfer lemma.

Cycle-1 exit checklist

- Collision equation separated from the conclusion it can actually prove.
- Unit-distance output supply distinguished from polynomial support compression.
- Directness generalized quantitatively to bounded multiplicity.
- Coordinate maps required to report domains, fibers, and preserved operations.
- Bounded distortion converted into an explicit polynomial output-container bound.
- Every high-dimensional claim reduced to a signed exponential-rate comparison.
- Five failures retained rather than deleted.
- One precise bottleneck selected for the next cycle.

[ABSTRACTED PRINCIPLE] Phase B Cycle 1 succeeds as a framework-building cycle: it can take a polynomial as input, expose its collision equation and normal forms, select a candidate mechanism, demand a support certificate, and run a gain-versus-cost audit. It does not yet supply a general theorem for rank-two mixed polynomials.

References and source map

1. Bloom, T. F.; Sawin, W.; Schildkraut, C.; Zhelezov, D. The sum-product conjecture is false for real numbers. arXiv:2605.28781v1 (2026). Used through the frozen Phase A reconstruction.

2. Phase A - Reproducing the High-Degree Sum-Product Construction. Final frozen version dated 26 August 2026. Controlling source for A=GP, its directness, bounded distortion, and parameter ledger.

3. OpenAI. Planar Point Sets with Many Unit Distances. Supplied proof manuscript. Proposition 2.2, Theorem 2.3, and Lemmas 2.4-2.6 control the source claims in Section 2 of this report.

4. Alon, N.; Bloom, T. F.; Gowers, W. T.; Litt, D.; Sawin, W.; Shankar, A.; Tsimerman, J.; Wang, V.; Wood, M. M. Remarks on the Disproof of the Unit Distance Conjecture. Supplied supporting manuscript.

5. The Unit Distance Problem and the Sum-Product Phenomenon: A Conceptual and Proof-Level Guide. Final frozen supplied guide. Used for source cross-checking and epistemic cautions.

6. unit-distance-cot.pdf. Supplied discovery-stage record. Not used as proof authority.

No external theorem imported
The additive/multiplicative normal-form statements in this report are elementary sufficient conditions. No classification theorem claiming that they exhaust all relevant polynomials is used.